

Financial Freedom Planning Model

1. input variables

a. user inputs

- demographic
 - + birthYear
 - + lifeExpectancy (user-defined or estimated using the estimation model)
 - + ffpAge (for scenarios 1, 3, 4)
- financial
 - + currentSavings
 - + desiredAnnualSpending (for scenarios 1, 2, 4)
 - + portfolio allocation (before and after FFP)
 - proportion of risky asset (u)
 - expected annual return rate of risky asset (mu)
 - risk-free annual return rate (r_f)

<https://www.britannica.com/money/modern-portfolio-theory-explained>

b. derived variables

- *currentAge* = *currentYear* - *birthYear*
- *retirementDuration*: $n = \text{lifeExpectancy} - \text{ffpAge}$ (how long money last?)

c. constraint

- $n \geq 0$
- $50 \leq \text{lifeExpectancy} \leq 100$

2. stage-based savings model (annual cash flow before FFP)

- for each stage k , k is in $[1, 6]$
 - + startAge_ k (fixed)
 - + endAge_ k (fixed)
 - + initial annual saving S_k
 - + annual growth rate g_k
- constraint:
 - + endAge_ k = startAge_ $(k+1)$ -> continuity
- annual saving at age t :

$$S(t) = S_k \cdot (1 + g_k)^{(t - \text{startAge}_k)}, \quad t \in [\text{startAge}_k, \text{endAge}_k)$$

- + notes
 - + includes all pre-FFP income (salary, rental, ...)
 - + stage boundaries are fixed inputs
 - + if no stages are provided => assume a single stage

3. asset modal (only passive income after FFP)

- for each asset j :
 - + type_j in {rental, pension}
 - + $I_j(\text{ffpAge})$: initial income of asset j at FFP
 - + r_j : growth rate
- for $t \geq \text{ffpAge}$, income of asset j :

$$I_j(t) = I_j(\text{ffpAge}) \cdot (1 + r_j)^{(t - \text{ffpAge})}$$

- +
 - note
 - + assets are not included in wealth accumulation
 - + rental income before FFP is included in $S(t)$
 - + income is used for consumption, not reinvested
 - + portfolio and income assets are **mutually exclusive**

4. wealth accumulation model (capital stock before FFP)

- initialize: $W_0 = W(\text{currentAge}) = \text{currentSavings}$
- expected portfolio return rate: $\rightarrow$ Investments growth

$$r_{\text{portfolio}} = u \cdot \mu + (1 - u) \cdot r_f$$

- +
 - + u : allocation of risky assets
 - + μ : expected annual return of risky assets
 - + r_f : risk-free asset
- pre-FFP wealth evolution:
 - +
 - $W(t+1) = W(t) \cdot (1 + r_{\text{savings}}) + S(t)$
 - + $W(t)$: accumulated wealth at age t
 - + r_{savings} : annual interest rate (bank ROI)
 - + $S(t)$: annual saving at age t
- post-FFP wealth evolution:
 - + after reach FFP, wealth continues to evolve under the user chosen return rate

$$W(t+1) = W(t) \cdot (1 + r_{\text{post}}) - E + \sum_j I_j(t)$$

- + E : annual spending
- + $I_j(t)$: income from assets
- note
 - + all income before FFP is already included in $S(t)$

- + For r_{saving} and r_{post} , there 2 options to collect this:

Option 1: Let user input r_{post} , r_{saving} (simple but less realistic)

Option2: Use portfolio theory for both (more realistic) → probably let user input u_{pre} , u_{post} (Intuition: Pre FFP: higher risky tolerance, post FFP: lower tolerance)

5. required wealth

- ensure funds last for the retirement duration (*If I withdraw **E every year**, and my money keeps growing at **r_{post}** , how big must my starting wealth be?*)
- formular

$$W_{\text{required}} = E \cdot \frac{1 - (1 + r_{\text{post}})^{-n}}{r_{\text{post}}}$$

+

If $r_{\text{post}} = 0$:

$$W_{\text{required}} = E \cdot n$$

+

- + E: desired annual spending
- + r_{post} : post-FFP interest rate (bank ROI)
- + n: retirement duration

- interpretation
 - + wealth continues earning interest after FFP
 - + spending is sustained over time

6. available spending at FFP

- total available spending consists of
 - + withdrawal from wealth
 - + passive income from assets
- wealth-based spending
 - + $W(\text{ffpAge})$ in this case will include all user savings (salary, income,...)

$$W(t + 1) = W(t) \cdot (1 + r_{\text{savings}}) + S(t)$$

+

$$E_{\text{wealth}} = W(\text{ffpAge}) \cdot \frac{r_{\text{post}}}{1 - (1 + r_{\text{post}})^{-n}}$$

+

- total available spending

$$E_{\text{available}} = E_{\text{wealth}} + \sum_j I_j(\text{ffpAge})$$

7. scenario formulation

- scenario 1: can I reach my FFP goal? - $W(\text{ffpAge}) \geq W_{\text{required}}$
 - + input: ffpAge, E

$$W(\text{ffpAge}) \geq E \cdot \frac{1 - (1 + r_{\text{post}})^{-n}}{r_{\text{post}}}$$

- + condition:

- scenario 2: when will I reach my FFP goal?

- + input: E
- + solve:

$$\text{Find } t \text{ such that } W(t) \geq E \cdot \frac{1 - (1 + r_{\text{post}})^{-(\text{lifeExpectancy} - t)}}{r_{\text{post}}}$$

- scenario 3: how much I can spend at FFP?

- + input: ffpAge
- + solve:

$$E = W(\text{ffpAge}) \cdot \frac{r_{\text{post}}}{1 - (1 + r_{\text{post}})^{-n}} + \sum_j I_j(\text{ffpAge})$$

- scenario 4: how much I should save each year to reach FFP goal?

- + input: ffpAge, E
- + objective

- determine a constant annual saving S^* (assume growth rate = 0) applied over the remaining pre-FFP period such that

$$W(\text{ffpAge}) = E \cdot \frac{1 - (1 + r_{\text{post}})^{-n}}{r_{\text{post}}}$$

- + stage handling

- let the user-defined stages cover ages up to endAge_k
- case 1: endAge_k < ffpAge
 - + an additional terminal stage k + 1 is introduced
 - startAge_(k+1) = endAge_k

- $\text{endAge}_{(k+1)} = \text{ffpAge}$
- $\text{growthRate } g_{(k+1)} = 0$
- $\text{annualSaving} = S^*$
- case 2: no stages provided
 - + entire period $[\text{currentAge}, \text{ffpAge})$ is treated as a single stage
 - + $S(t) = S^*$
- case 3: stages already cover up to FFP
 - + not handle $\Rightarrow$ prompt user to select other scenarios or edit ffpAge or endAge_k so that $\text{engAge}_k < \text{ffpAge}$
- + solution method
 - binary search ($W(\text{ffpAge})$ depends recursively on savings $\Rightarrow$ can not derive saving S^* directly)

Algorithm

1. Define search bounds:

- $S_{\min}^* = 0$
- $S_{\max}^* = \text{sufficiently large value}$

2. Iterate:

- Let $S^* = \frac{S_{\min}^* + S_{\max}^*}{2}$
- Simulate wealth accumulation using:

$$W(t+1) = W(t)(1 + r_{\text{savings}}) + S(t)$$

- Compare $W(\text{ffpAge})$ with W_{required}

3. Update bounds:

- If $W(\text{ffpAge}) < W_{\text{required}}$:
 $\rightarrow$ increase S^*
- Otherwise:
 $\rightarrow$ decrease S^*

4. Repeat until convergence



8. notes

- Pre-FFP portfolio: applies only to financial assets (cash, savings, stocks, bonds); income (e.g., rental) is included in $S(t)$, not portfolio
- Post-FFP portfolio: applies only to remaining financial wealth; excludes rental and pension, which are modeled separately as passive income
- Portfolio allocation UI hint
 - + pre-FFP: "Applies to your savings and investments only. Income (e.g., rental) is already included in your life-stage savings." at step 2 j dok

- + post-FFP: “Applies to your remaining savings after retirement. Rental income and pension are counted separately.” at step 3 j dok